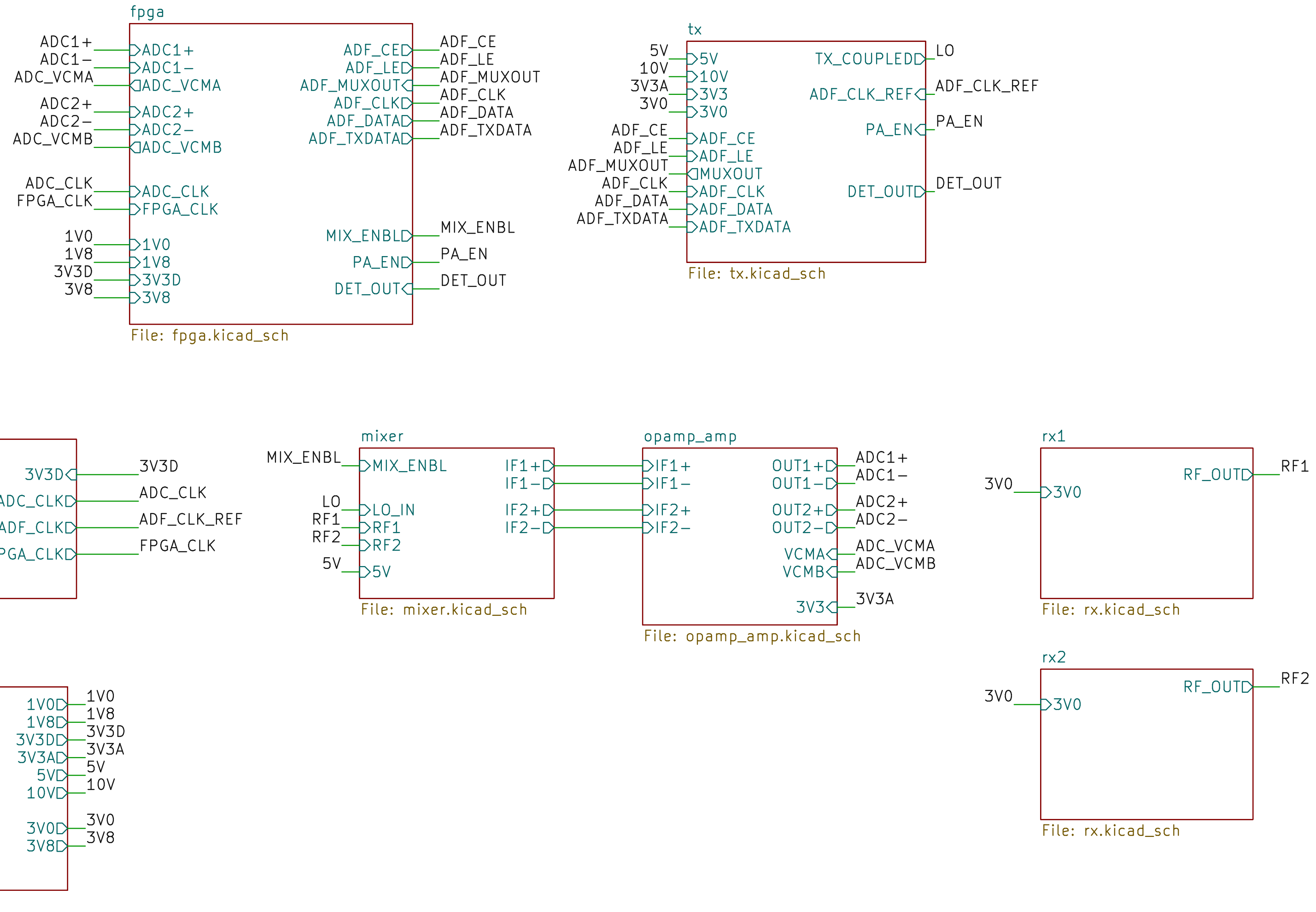
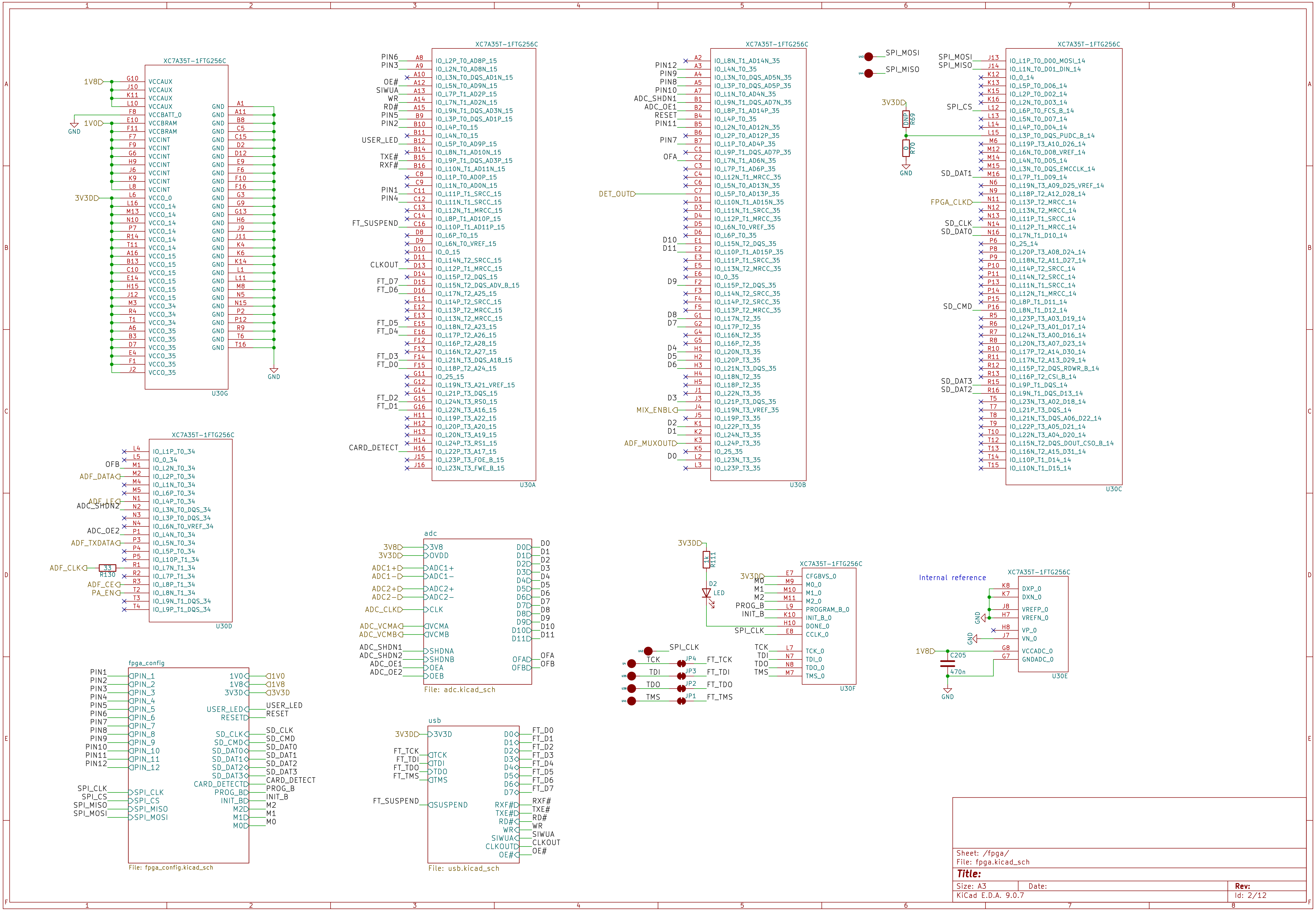


Sheet: /
File: fmcw3.kicad_sch

Title: FMCW3 by CENK KESKİN

Size: A4 | Date: | Rev:
KiCad E.D.A. 9.0.7 | Id: 1/12



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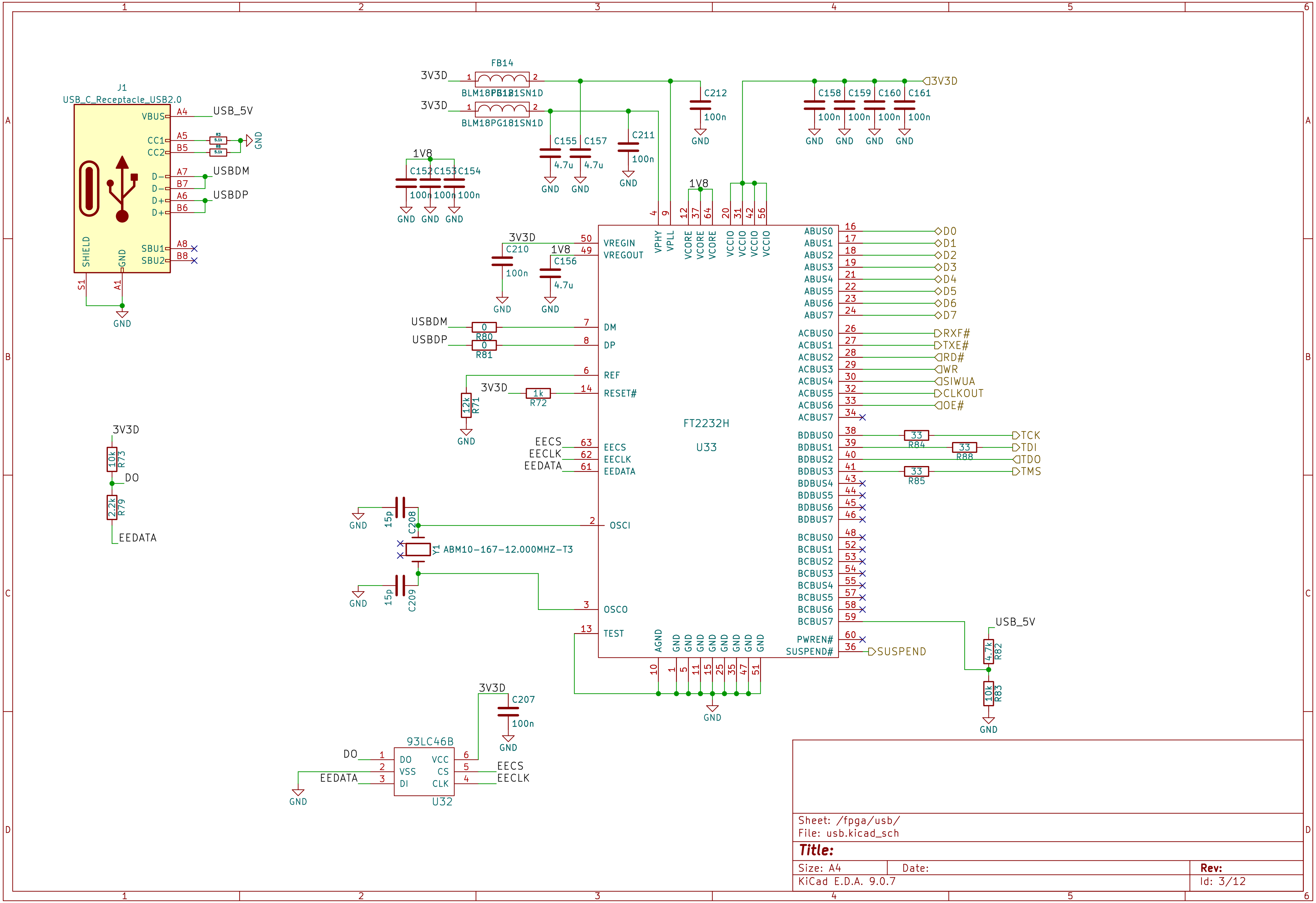
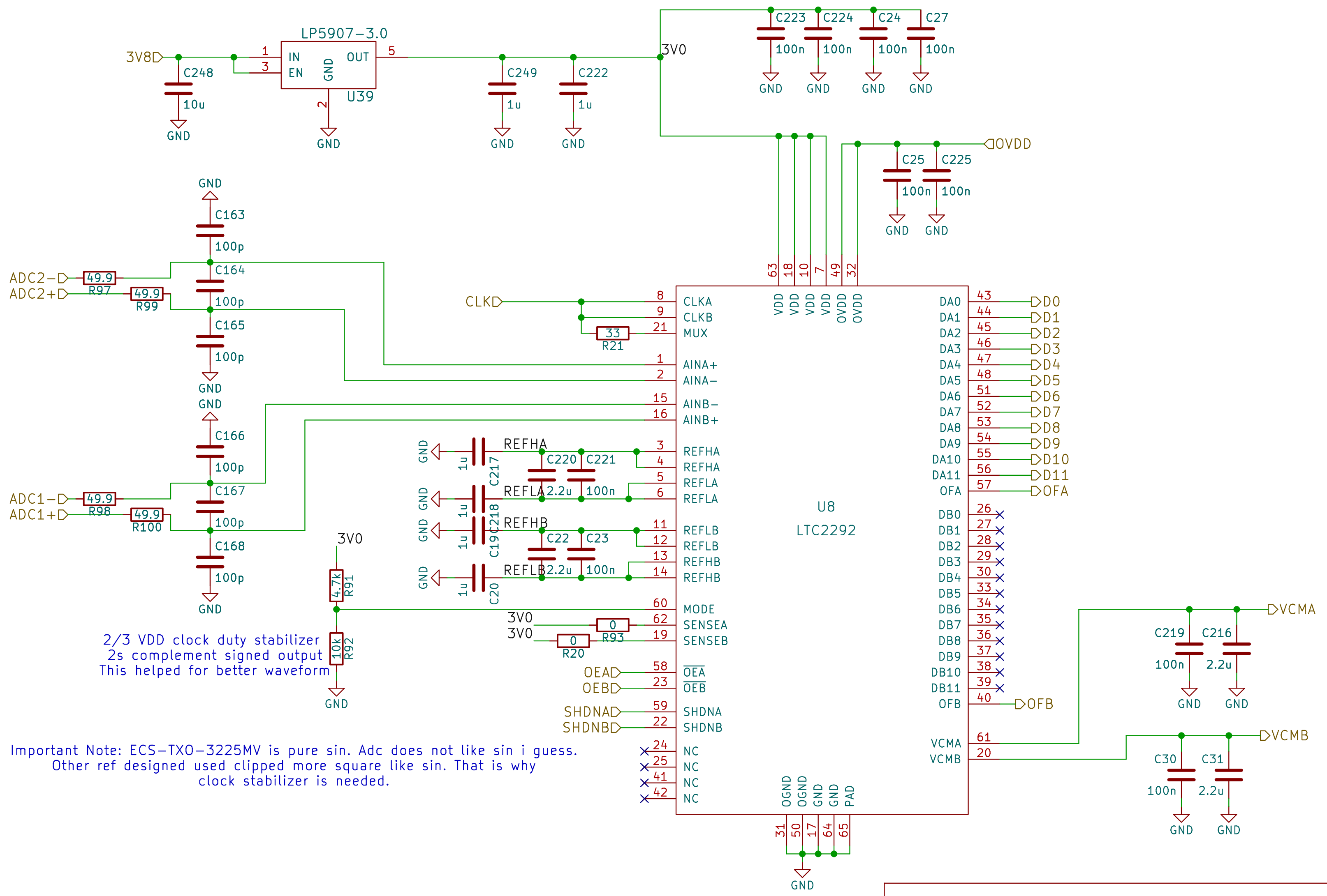
File: fpga.kicad_sch

Title:

Size: A3 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 2/12

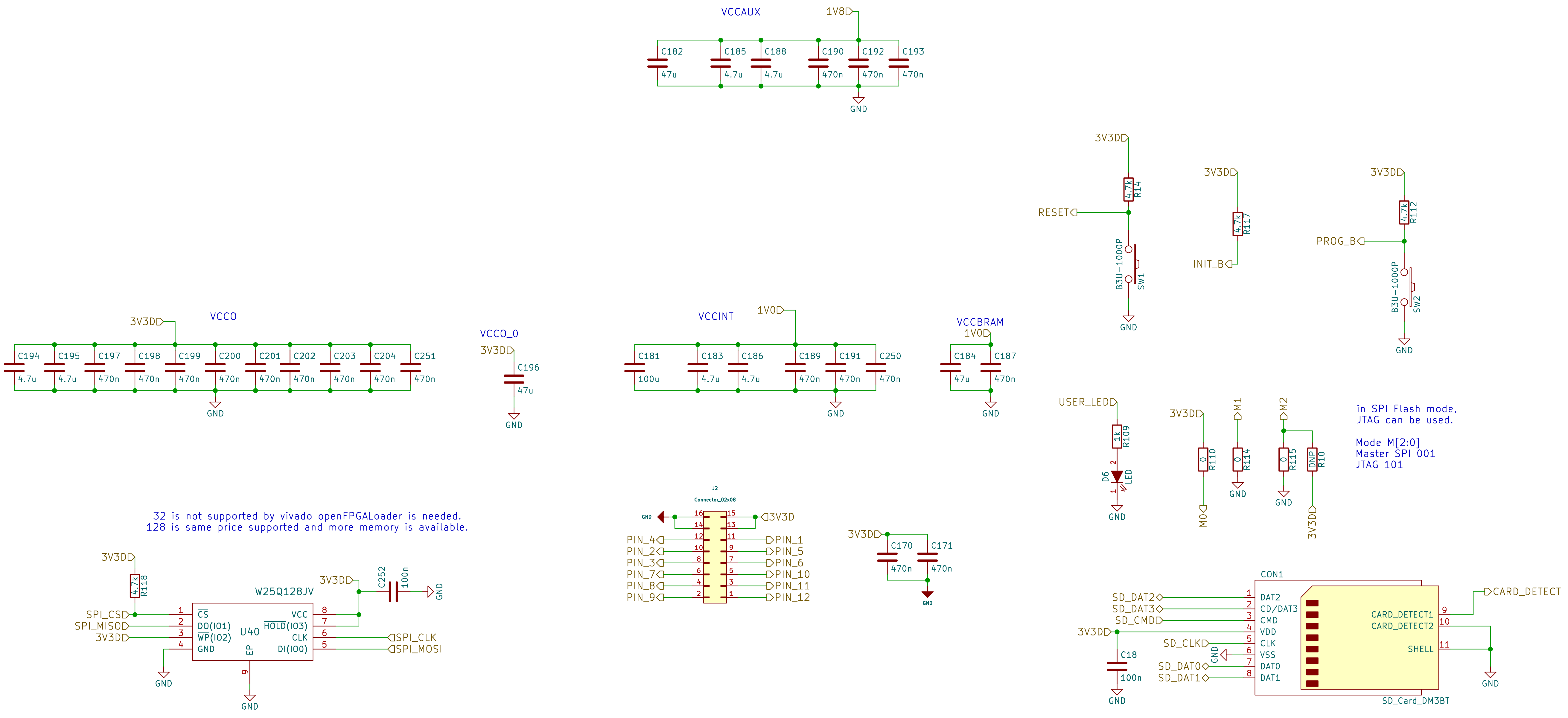

J1
USB_C_Receptacle_USB2.0
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
GND
A4
A5
B5
A7
B7
A6
B6
A8
B8
S1
A1
USB_5V
USBDM
USBDP
GND
FB14
3V3D
BLM18PG181SN1D
C212
100n
C158
C159
C160
C161
100n
3V3D
1V8
C152
C153
C154
C155
C157
4.7u
C211
C210
C156
VREGIN
VREGOUT
VPHY
VPLL
VCORE
VCCIO
ABUS0
ABUS1
ABUS2
ABUS3
ABUS4
ABUS5
ABUS6
ABUS7
D0
D1
D2
D3
D4
D5
D6
D7
ACBUS0
ACBUS1
ACBUS2
ACBUS3
ACBUS4
ACBUS5
ACBUS6
ACBUS7
RXF#
TXE#
RD#
WR
SIWUA
CLKOUT
OE#
BDBUS0
BDBUS1
BDBUS2
BDBUS3
BDBUS4
BDBUS5
BDBUS6
BDBUS7
TCK
TDI
TDO
TMS
R84
R88
R85
33
BCBUS0
BCBUS1
BCBUS2
BCBUS3
BCBUS4
BCBUS5
BCBUS6
BCBUS7
PWREN#
SUSPEND#
SUSPEND
DM
DP
REF
RESET#
EECS
EECLK
EEDATA
OSCI
OSCO
TEST
AGND
FT2232H
U33
R80
R81
0
R71
12k
R72
1k
R73
10k
DO
R79
2.2k
C208
C209
15p
Y1 ABM10-167-12.000MHZ-T3
R82
4.7k
R83
10k
C207
93LC46B
U32
DO
VSS
DI
VCC
CS
CLK
Sheet: /fpga/usb/
File: usb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 3/12



Sheet: /fpga/adc/
File: adc.kicad_sch

Title:

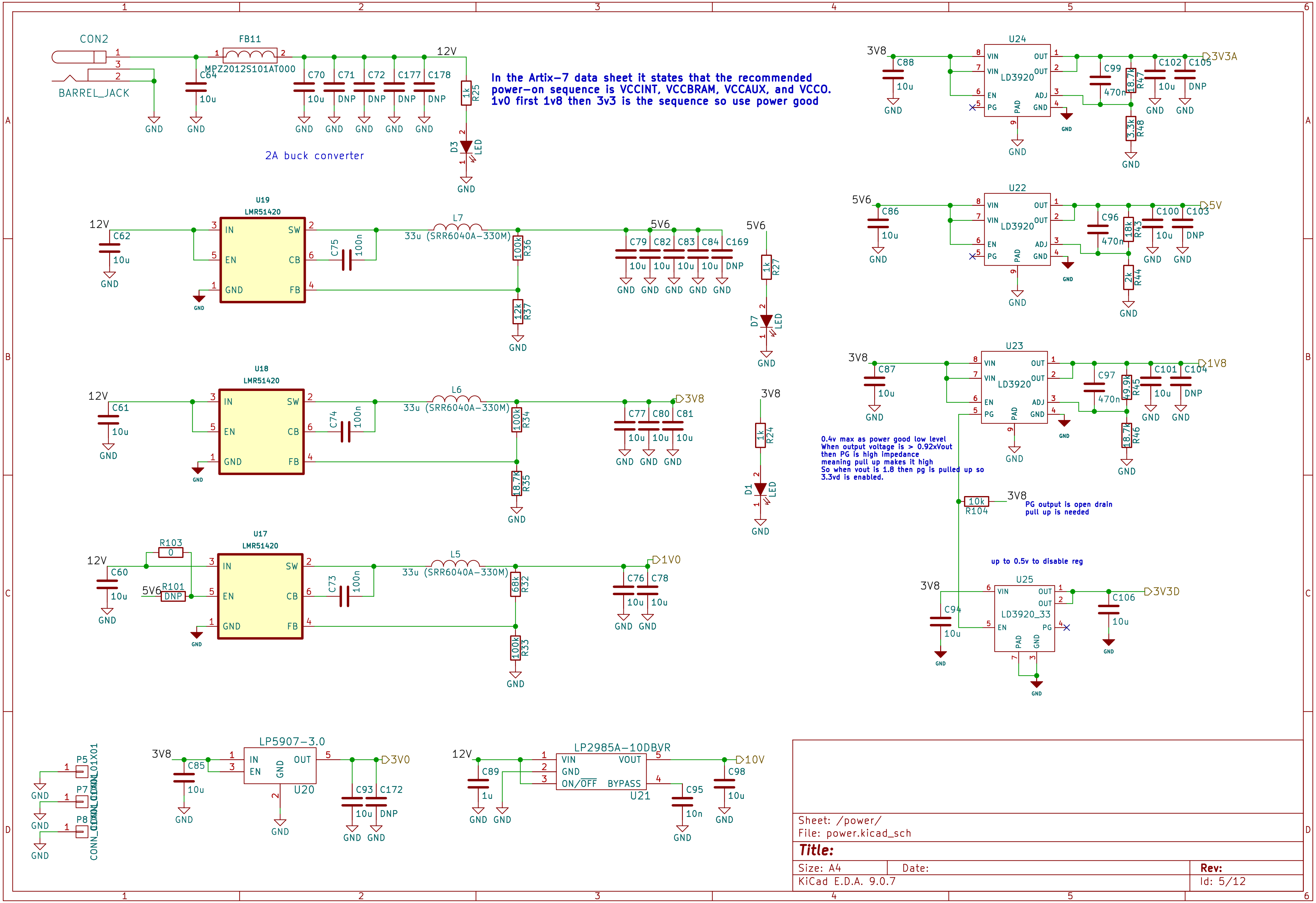
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 4/12



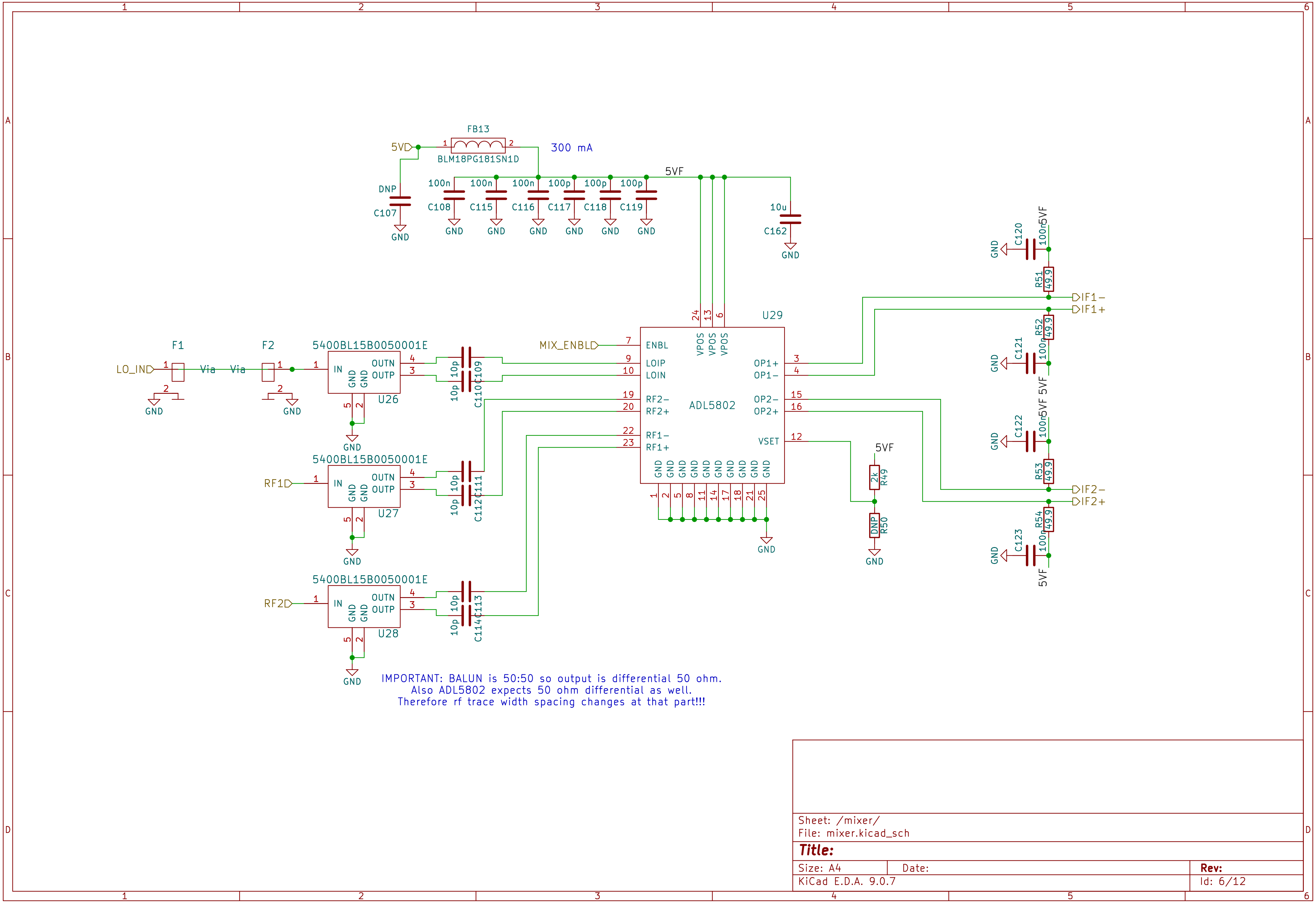
Sheet: /fpga/fpga_config/
File: fpga_config.kicad_sch

Title:

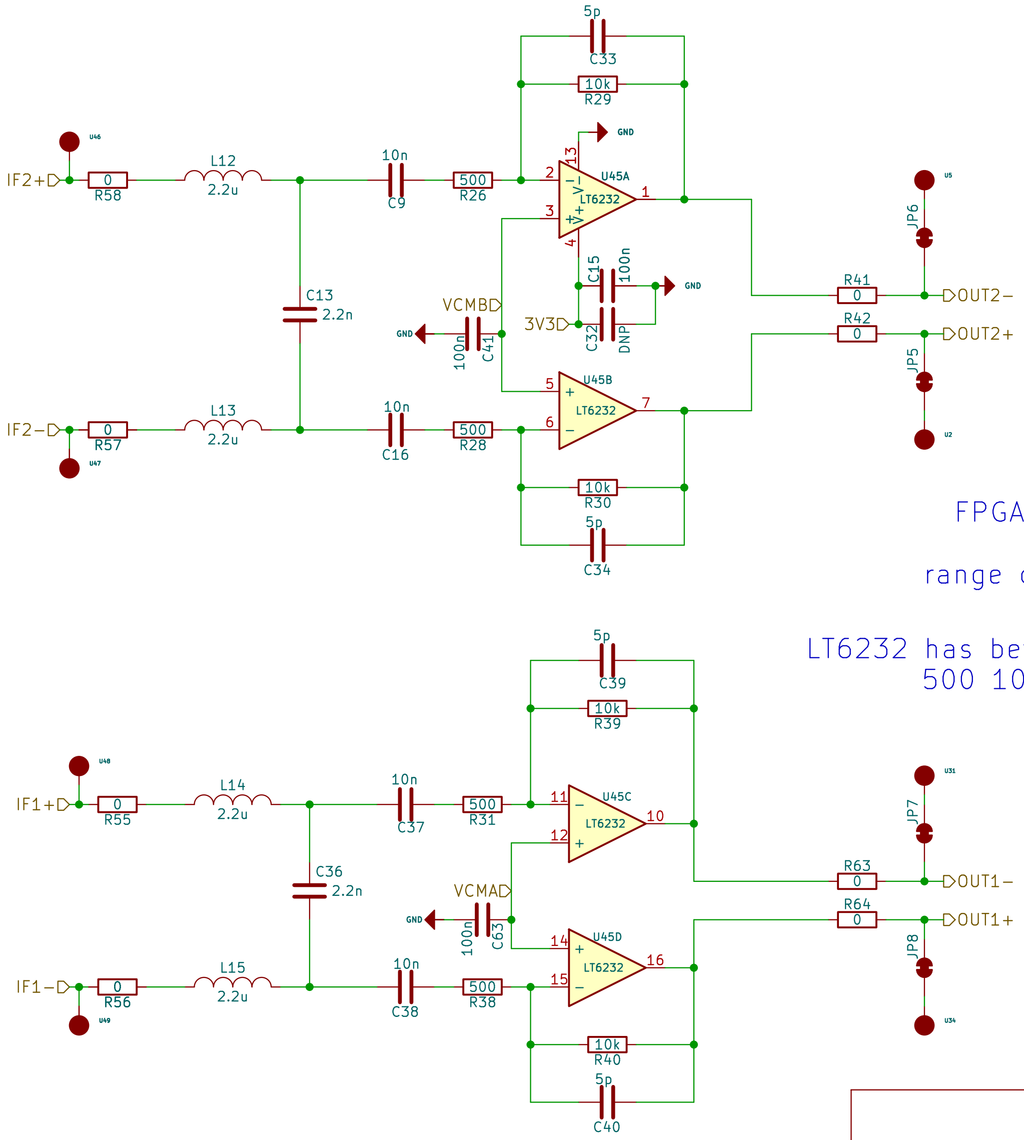
Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 12/12



CON2
BARREL_JACK
FB11
MPZ2012S101AT000
12V
C64 10u
C70 10u
C71 DNP
C72 DNP
C177 DNP
C178 DNP
R25 1k
D3 LED
GND
2A buck converter
In the Artix-7 data sheet it states that the recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO.
1v0 first 1v8 then 3v3 is the sequence so use power good
U19 LMR51420
IN EN GND SW CB FB
L7 33u (SRR6040A-330M)
C62 10u
C75 100n
R36 100k
R37 12k
5V6
C79 10u
C82 10u
C83 10u
C84 10u
C169 DNP
R27 1k
D7 LED
U18 LMR51420
L6 33u (SRR6040A-330M)
C61 10u
C74 100n
R34 100k
R35 18.7k
3V8
C77 10u
C80 10u
C81 10u
R24 1k
D1 LED
U17 LMR51420
R103 0
R101 DNP
L5 33u (SRR6040A-330M)
C60 10u
C73 100n
R32 68k
R33 100k
1V0
C76 10u
C78 10u
U24 LD3920
VIN OUT EN ADJ PG PAD GND
C88 10u
C99 470n
R47 18.7k
R48 3.3k
C102 10u
C105 DNP
3V3A
U22 LD3920
5V
C86 10u
C96 470n
R43 18k
R44 2k
C100 10u
C103 DNP
U23 LD3920
1V8
C87 10u
C97 470n
R45 49.9k
R46 18.7k
C101 10u
C104 DNP
0.4v max as power good low level
When output voltage is > 0.92xVout
then PG is high impedance
meaning pull up makes it high
So when vout is 1.8 then pg is pulled up so
3.3vd is enabled.
R104 10k
PG output is open drain
pull up is needed
up to 0.5v to disable reg
U25 LD3920_33
C94 10u
C106 10u
3V3D
P5
P7
P8
CONN_01X01
LP5907-3.0
U20
IN EN GND OUT
C85 10u
C93 10u
C172 DNP
3V0
LP2985A-10DBVR
U21
VIN GND ON/OFF VOUT BYPASS
C89 1u
C95 10n
C98 10u
10V
Sheet: /power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 5/12



FB13
BLM18PG181SN1D
300 mA
5VD
5VF
DNP
C107
100n C108
100n C115
100n C116
100p C117
100p C118
100p C119
10u C162
GND
U29
ADL5802
ENBL LOIP LOIN RF2− RF2+ RF1− RF1+
VPOS VPOS VPOS
OP1+ OP1− OP2− OP2+ VSET
MIX_ENBLD
LO_IND
F1
F2
Via Via
5400BL15B0050001E
U26
U27
U28
IN OUTN OUTP GND GND
RF1D
RF2D
10p C109 C110 C111 C112 C113 C114
C120 C121 C122 C123 100n
R51 R52 R53 R54 49.9
R49 2k
R50 DNP
IF1− IF1+ IF2− IF2+
IMPORTANT: BALUN is 50:50 so output is differential 50 ohm.
Also ADL5802 expects 50 ohm differential as well.
Therefore rf trace width spacing changes at that part!!!
Sheet: /mixer/
File: mixer.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 6/12



FPGA FIR Filter works perfectly.
Second stage for HPF
range compensation filer is removed

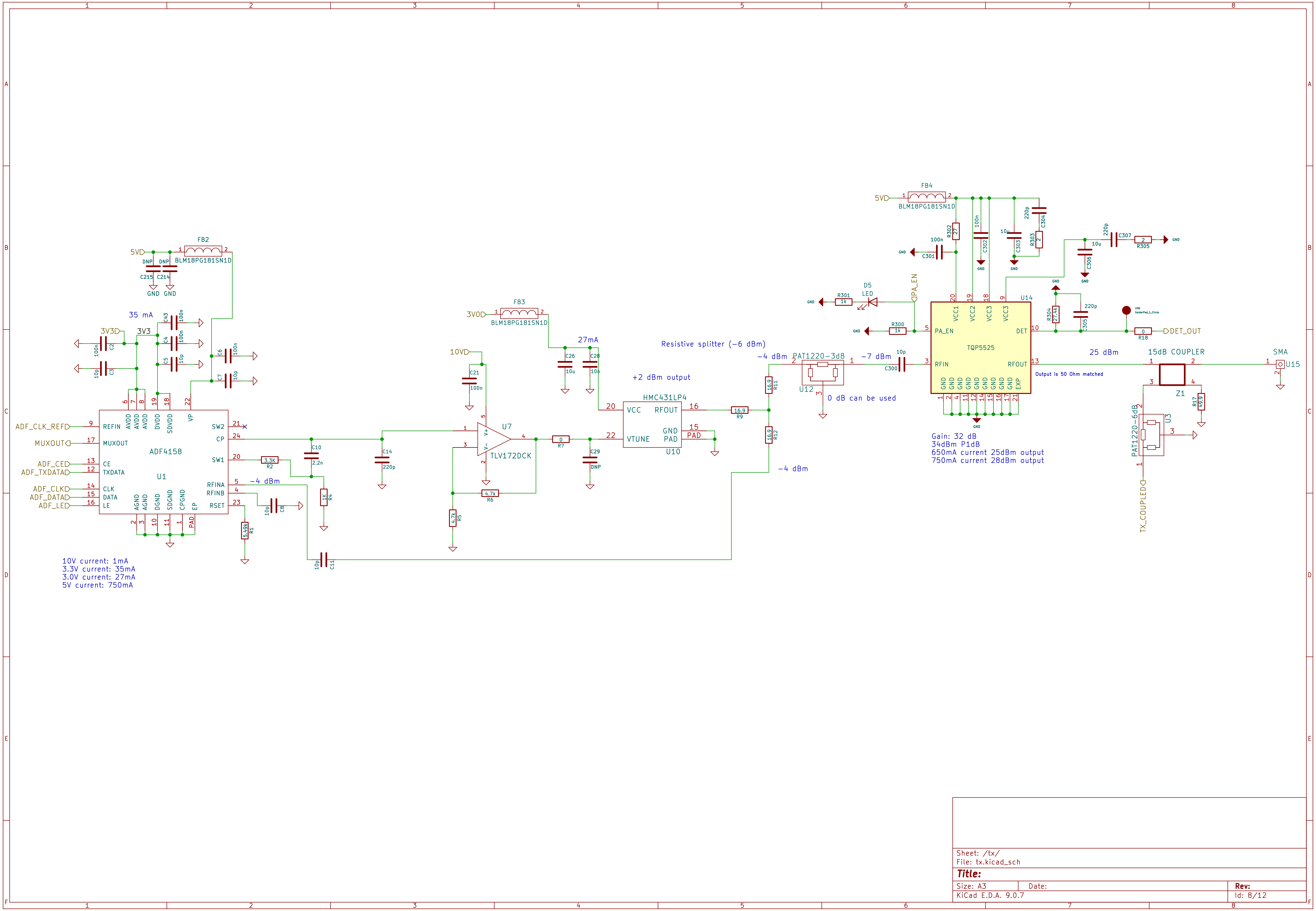
LT6232 has better noise performance as 1.1nV/√Hz
500 10K has less noise contribution

Sheet: /opamp_amp/
File: opamp_amp.kicad_sch

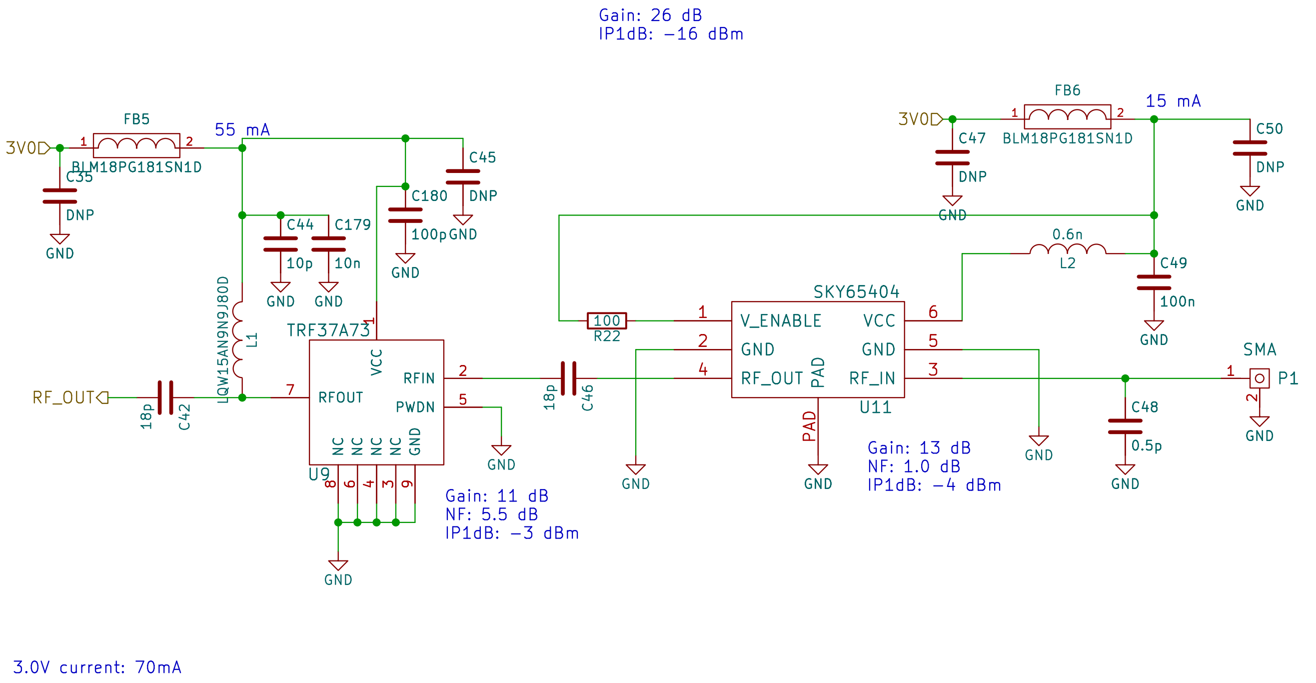
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.7 | Id: 7/12



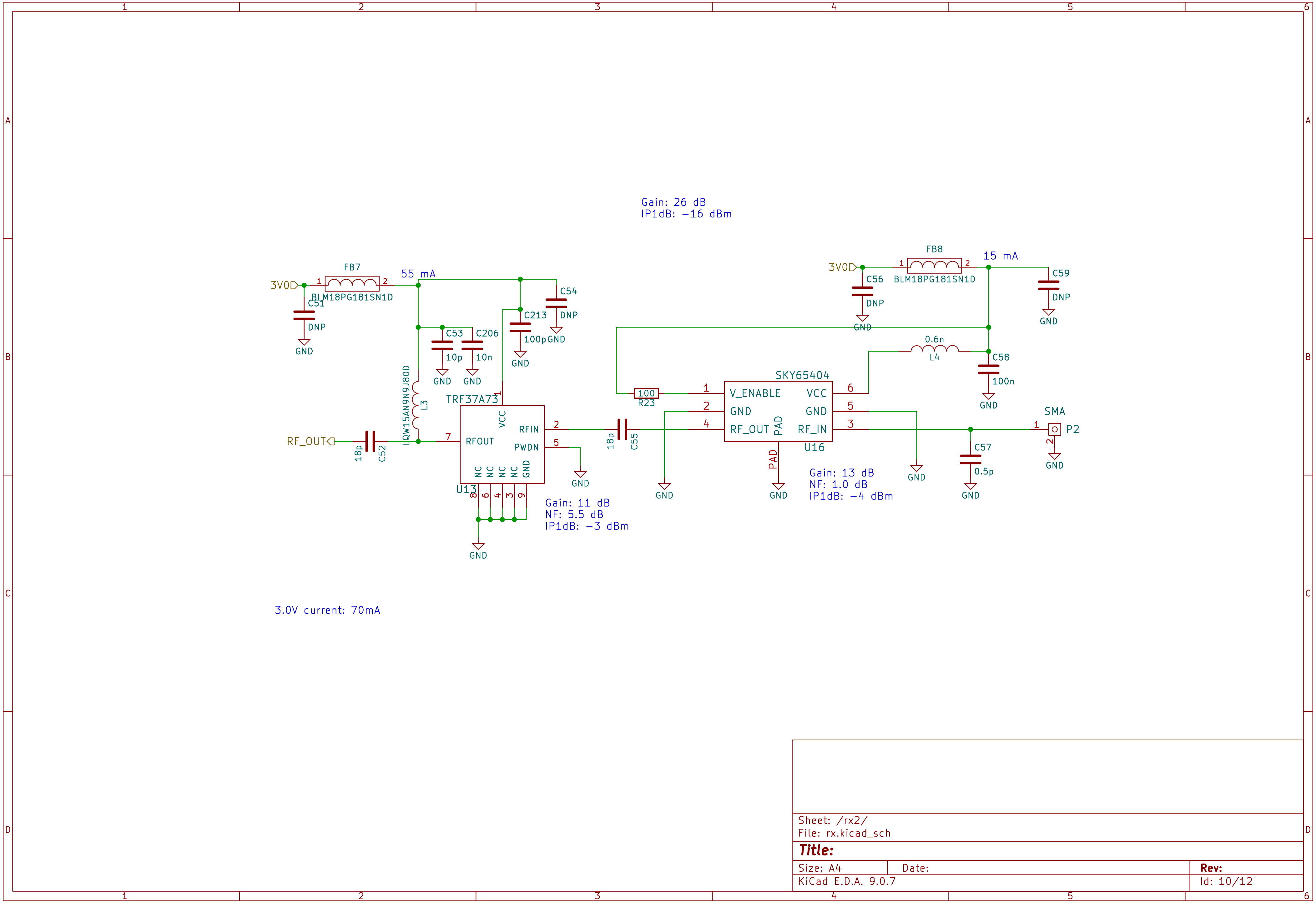
ADF4158
U1
35 mA
3V3
FB2
BLM18PG181SN1D
5V
DNP
GND
ADF_CLK_REFD
MUXOUT
ADF_CED
ADF_TXDATAD
ADF_CLKD
ADF_DATAD
ADF_LED
REFIN
CE
TXDATA
CLK
DATA
LE
AVDD
DVDD
SDVDD
VP
SW2
CP
SW1
RFINA
RFINB
RSET
-4 dBm
10V current: 1mA
3.3V current: 35mA
3.0V current: 27mA
5V current: 750mA
FB3
3V0
10V
U7
TLV172DCK
27mA
HMC431LP4
U10
VCC
RFOUT
VTUNE
GND
PAD
+2 dBm output
Resistive splitter (-6 dBm)
-4 dBm
PAT1220-3dB
U12
-7 dBm
0 dB can be used
FB4
D5
LED
PA_EN
U14
TQP5525
RFIN
DET
VCC1
VCC2
VCC3
EXP
Output is 50 Ohm matched
25 dBm
15dB COUPLER
Z1
SMA
U15
PAT1220-6dB
U3
DET_OUT
TX_COUPLED
Gain: 32 dB
34dBm P1dB
650mA current 25dBm output
750mA current 28dBm output
Sheet: /tx/
File: tx.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 8/12



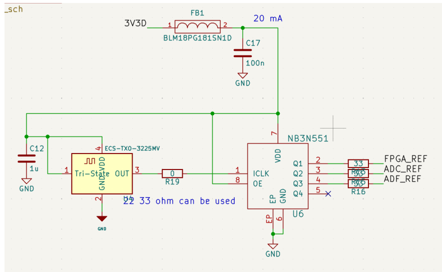
Sheet: /rx1/
File: rx.kicad_sch

Title:

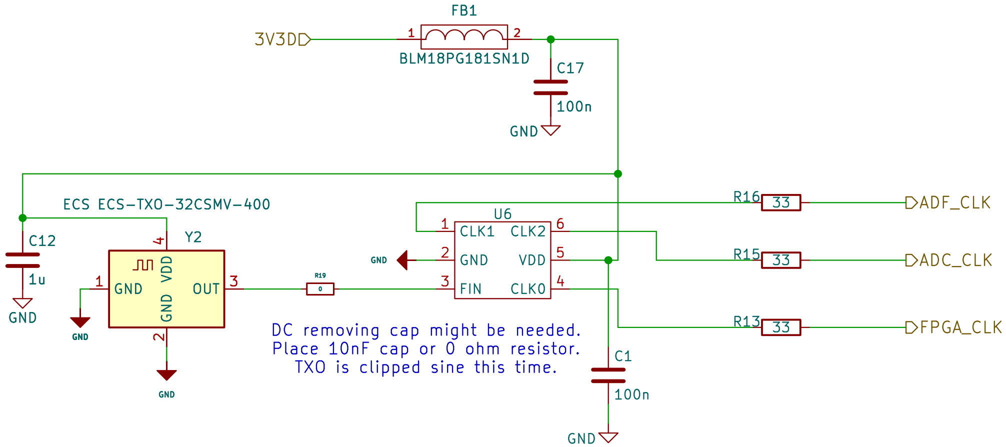
Size: A4
Date:
KiCad E.D.A. 9.0.7
Rev:
Id: 9/12



Gain: 26 dB
IP1dB: −16 dBm
3V0D
FB7
BLM18PG181SN1D
55 mA
C51
DNP
C54
DNP
C213
100p
C53
10p
C206
10n
LQW15AN9N9J80D
L3
TRF37A73
VCC
RFIN
RFOUT
PWDN
NC NC NC NC GND
U13
RF_OUT
18p
C52
18p
C55
Gain: 11 dB
NF: 5.5 dB
IP1dB: −3 dBm
3V0D
FB8
BLM18PG181SN1D
15 mA
C56
DNP
C59
DNP
0.6n
L4
C58
100n
100
R23
SKY65404
V_ENABLE
VCC
GND
GND
RF_OUT
RF_IN
PAD
U16
SMA
P2
C57
0.5p
Gain: 13 dB
NF: 1.0 dB
IP1dB: −4 dBm
GND
3.0V current: 70mA
Sheet: /rx2/
File: rx.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 10/12



This is the prev design:
NB3N551 is obs.
Also this txo is generating pure sin.
ADC might need cmos like input not sin.
That could be the reason of duty stabilizer.



DC removing cap might be needed.
Place 10nF cap or 0 ohm resistor.
TXO is clipped sine this time.

Sheet: /clock/		
File: clock.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 11/12